

Title

Secondary outcomes: Craving reduction and biomarker changes in stimulated versus control participants

Abstract

Secondary analysis examined mechanistic and exploratory outcomes from the 24-week neuromodulation trial. At endpoint, the active stimulation group (versus sham) demonstrated lower self-reported craving intensity (mean difference -0.36, 95% CI -0.65 to -0.07, $p=0.016$) measured on a validated rating scale. Serum carbohydrate deficient transferrin, a marker of recent heavy drinking, showed greater reduction in the stimulation group (-0.33 units, 95% CI -0.65 to -0.01, $p=0.045$). Quality-of-life scores and mood assessments showed non-significant trends favoring active treatment. Cognitive testing revealed no between-group differences. Adverse events occurred similarly in both arms (41% active vs 37% sham), with most being mild. Data completeness was affected by losses to follow-up and incomplete diary entries. These findings suggest neuromodulation may influence psychological drive to drink and reflect reduction in heavy consumption days, though further investigation of mechanisms is warranted.